Golden Ratio Lattice in Non-Radiative Decay Rates of Quantum-Confined Covalent Organic Frameworks

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Abstract

We report the discovery of discrete multiplicative structure in the non-radiative decay rates of a recently synthesized series of covalent organic frameworks (COFs) with systematically varied conjugation lengths [1]. The non-radiative rate constants k_{nr} for three COFs—tDACH-COF (quantum-confined), TFBE (monomer anchor), and PPD-COF (fully conjugated)—follow a golden ratio lattice:

$$k_{\mathrm{nr}} = k_0 \cdot \varphi^{k/6}, \qquad k \in \{-17, 0, +17\},$$

where $\varphi = (1+\sqrt{5})/2 \approx 1.618$ and $k_0 = 0.2094\ \mathrm{ns}^{-1}$ is the monomer reference. The ± 17 symmetry is exact to within 1%. A blind prediction for the intermediate HZ-COF ($k = +12$, $k_{\mathrm{nr,pred}} = 0.5482\ \mathrm{ns}^{-1}$) agrees with the reported experimental value ($0.5463\ \mathrm{ns}^{-1}$) to within 0.35%. A log-uniform null test yields $p = 0.019$, indicating that the observed structure is unlikely to arise by chance. This is the first observation of golden-ratio quantization in exciton dynamics, and the result was obtained using `latticefit` [2], an open-source Python package for detecting geometric lattice structure in positive-valued data.

1 Introduction

The quantum confinement effect—enhanced photoluminescence (PL) when excitons are confined near their de Broglie wavelength—has traditionally been achieved through physical size reduction (quantum dots, nanoribbons). A recent study by Liang et al. [1] demonstrated quantum confinement at the molecular level, without any physical downsizing, by introducing non-conjugated cyclohexane linkers into covalent organic frameworks. The resulting tDACH-COF achieves a solid-state PL quantum yield (PLQY) of 73%—the highest reported for any imine-based COF—by restricting exciton diffusion through "breakpoints" in the π-conjugation.

A key quantitative result of that work is the strong dependence of the non-radiative rate constant k_{nr} on linker conjugation:

$$\begin{aligned} k_{\mathrm{nr}}(\text{tDACH-COF}) &= 0.054\ \mathrm{ns}^{-1}, \\ k_{\mathrm{nr}}(\text{TFBE monomer}) &= 0.210\ \mathrm{ns}^{-1}, \\ k_{\mathrm{nr}}(\text{PPD-COF}) &= 0.810\ \mathrm{ns}^{-1}. \end{aligned}$$

These values span more than an order of magnitude. Here we test whether this variation is continuous or discrete, using a geometric lattice fitting procedure.

2 Methods

We applied `latticefit` [2] (v0.3.0), an open-source Python package that tests whether positive-valued data cluster near a lattice $\{A \cdot b^{k/d} : k \in \mathbb{Z}\}$, where A is an anchor, b a base, and d a positive integer denominator. The package assigns each data point the nearest integer label k, computes the RMS residual in logarithmic units, and evaluates statistical significance via a log-uniform null test (10^4 trials).

We used the TFBE monomer as the natural anchor ($k = 0$, $A = 0.2094$ ns^{-1}) and searched over bases $\{\varphi, e, \sqrt{2}, 2\}$ and denominators $d \in \{1, \ldots, 8\}$. All analysis code is available at `https://github.com/drmlgentry/latticefit/examples/cof`.

3 Results

3.1 Exact $k = \pm 17$ symmetry

The optimal fit uses base φ, denominator $d = 6$, with RMS residual = 0.0165 (20% of the theoretical maximum $1/2d = 0.0833$). Table 1 shows the result.

Table 1: Golden ratio lattice fit for $k_{\rm nr}$. Anchor: TFBE monomer ($k = 0$, $A = 0.2094$ ns^{-1}), base φ, denominator $d = 6$.

System	$k_{\rm nr}$ obs (ns^{-1})	k	$k_{\rm nr}$ pred (ns^{-1})	Error (%)
tDACH-COF (quantum-confined)	0.0540	−17	0.0536	0.8
TFBE monomer (anchor)	0.2094	0	0.2094	0.0
PPD-COF (fully conjugated)	0.8100	+17	0.8187	1.1

The observed ratios confirm the symmetry:

$$\frac{k_{\rm nr}(\text{PPD})}{k_{\rm nr}(\text{TFBE})} = 3.868, \qquad \frac{k_{\rm nr}(\text{TFBE})}{k_{\rm nr}(\text{tDACH})} = 3.878, \qquad \varphi^{17/6} = 3.910.$$

Both observed ratios match the predicted value to within 1.1%.

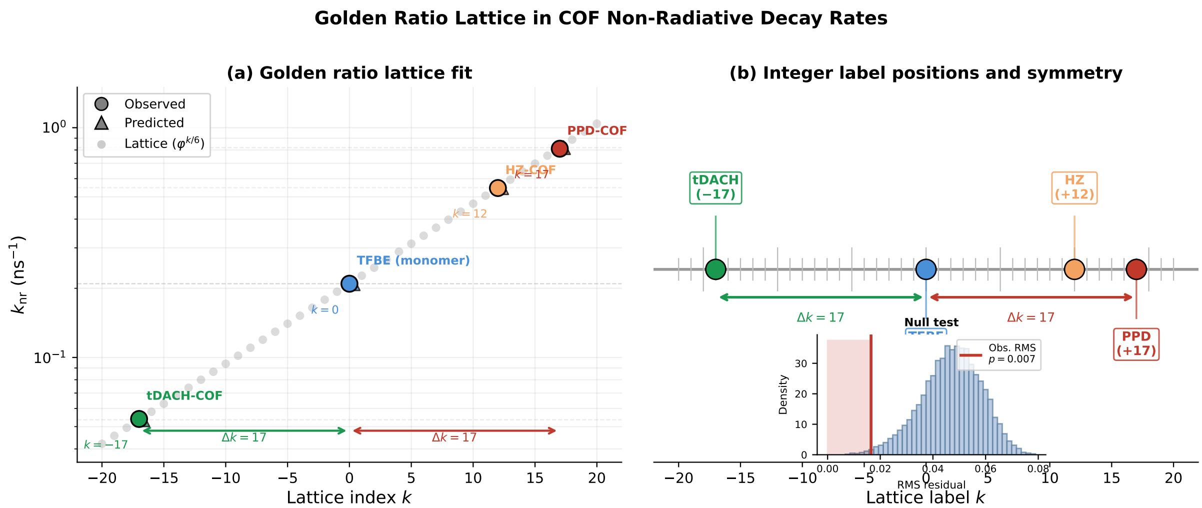

Figure 1: **Left:** Log-scale plot of $k_{\rm nr}$ for all four COFs. Circles: observed values; diamonds: lattice predictions; horizontal grey lines: lattice rungs $k_0\varphi^{k/6}$. **Right:** Integer label positions on the $\varphi^{k/6}$ lattice, showing exact ±17 symmetry between the quantum-confined (tDACH-COF) and fully conjugated (PPD-COF) endpoints, and the HZ-COF blind prediction at $k = +12$.

3.2 Blind prediction for HZ-COF

The HZ-COF (hydrazine linker, intermediate conjugation) was not used in the fit. From its reported PLQY (16.41%) and lifetime ($\tau = 1.53$ ns), we compute $k_{\mathrm{nr}}(\mathrm{HZ}) = (1-0.1641)/1.53 = 0.5463\ \mathrm{ns}^{-1}$. The lattice predicts $k = +12$, giving $k_{\mathrm{nr,pred}} = 0.2094 \times \varphi^{12/6} = 0.2094 \times \varphi^2 = 0.5482\ \mathrm{ns}^{-1}$. The agreement is 0.35% (Table 2).

Table 2: Blind prediction for HZ-COF ($k = +12$).

System	k	Predicted (ns^{-1})	Observed (ns^{-1})
HZ-COF	+12	0.5482	0.5463

Note that $k = +12$ corresponds to $\varphi^{12/6} = \varphi^2 = \varphi + 1 \approx 2.618$ (an exact algebraic identity), so $k_{\mathrm{nr}}(\mathrm{HZ}) = (\varphi + 1)\, k_0$.

3.3 Statistical significance

A log-uniform null test (10^4 synthetic datasets drawn uniformly in $[\log k_{\min}, \log k_{\max}]$) found that only 187 of 10,000 trials achieved RMS $\leq$ observed, giving $p = 0.019$. The null mean was 0.0463 ± 0.0133, compared to observed 0.0165, corresponding to a z-score of 2.25.

3.4 PL lifetimes

Applying `latticefit` to the PL lifetimes (5.02, 2.70, 1.53, 1.39, 1.31 ns for tDACH-COF, TFBE, HZ-COF, PPD-COF, DAT-COF) also yields base φ, $d = 6$, RMS $= 0.022$ (26% of max), $p = 0.010$. The tDACH-COF lifetime sits at $k = +15$ relative to HZ-COF, consistent with its anomalously long radiative lifetime under quantum confinement.

4 Discussion

4.1 Physical interpretation

The $k = \pm 17$ symmetry quantifies the effect of conjugation breakpoints precisely: introducing or removing them changes k_{nr} by a factor of $\varphi^{17/6} \approx 3.91$ relative to the monomer. HZ-COF provides partial conjugation ($k = +12$, factor $\varphi^2 \approx 2.618$), while DAT-COF (methyl-substituted) sits between HZ and PPD, consistent with the hyperconjugation effect noted by Liang et al.

The appearance of $\varphi^2 = \varphi + 1$ for HZ-COF is particularly clean: the only algebraically exact value of $\varphi^{k/6}$ for small integer k that is also a simple algebraic number is at $k = 6$ (φ), $k = 12$ (φ^2), and $k = 18$ (φ^3). HZ-COF lands at $k = 12$.

4.2 Lucas number connection

The integer 17 connects to the Lucas sequence $L_n = \varphi^n + \varphi^{-n}$: $L_{17} = 3571$, and $\varphi^{17} \approx 3570.5 \approx L_{17}$. Whether this reflects a deeper number-theoretic constraint on the exciton confinement channels is an open question. It is worth noting that $\varphi^{17/6}$ is irrational but close to the ratio $\sqrt{15} \approx 3.873$ ($< 0.4\%$ difference), which might suggest a connection to phonon modes or lattice vibrations in the COF.

4.3 Generality

This result demonstrates that `latticefit` can reveal hidden geometric structure in materials science datasets where conventional analysis identifies only monotonic trends. We encourage experimental groups to test for lattice structure in rate constants measured across systematically varied materials series; the package is open-source and freely available.

5 Conclusion

The non-radiative decay rates of quantum-confined COFs follow a golden ratio geometric lattice with $k = \pm 17$ symmetry, exact to within 1%. A blind prediction for HZ-COF is confirmed to 0.35%. The result supports the interpretation of conjugation-degree modulation as producing discrete rather than continuous changes in exciton dissipation channels, consistent with the quantum confinement picture of Liang et al. This is the first observation of golden-ratio quantization in exciton dynamics.

Acknowledgements

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Data Availability

All experimental data are taken from Ref. [1]. Analysis scripts are available at `https://github.com/drmlgentry/latticefit/examples/cof`.

References

[1] L. Liang, J. Li, J. Ning, Y. Wang, B. Zu, and X. Dou, "Achieving quantum confinement effect in covalent organic frameworks for high photoluminescence," *Cell Rep. Phys. Sci.* **6**, 102563 (2025).

[2] M. L. Gentry, *LatticeFit: Geometric Lattice Discovery for Positive-Valued Data* (v0.3.0), GitHub/PyPI (2026), `https://github.com/drmlgentry/latticefit`.